

Packet Tracer Office Network

Final Lab Report

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Item	Details
Project Folder	01-Networking / Packet-Tracer-Office-Network
Platform	Cisco Packet Tracer
Primary Devices	Cisco 2911 routers, Cisco 2960 switches, PCs, wireless access points
Network Design	Head Office and Branch Office enterprise-style network
Primary Skills	VLANs, trunking, router-on-a-stick, inter-VLAN routing, ACLs, DHCP/DNS planning, testing
Report Purpose	Document design, configuration, testing, troubleshooting, and final project outcome

Disclaimer

This report documents a Cisco Packet Tracer educational lab completed for training and portfolio development. All configurations and testing were performed in a simulated lab environment.

1. Executive Summary

This project involved designing and documenting a complete office network in Cisco Packet Tracer. The network connected a Head Office and Branch Office using Cisco routers and switches, department VLANs, router-on-a-stick inter-VLAN routing, trunk links, wireless access, guest segmentation, and access control planning.

The project demonstrates practical networking fundamentals required for IT support, network administration, cybersecurity lab work, and technical troubleshooting. It also includes lessons learned from configuration errors, especially the proper order of configuring 802.1Q encapsulation before assigning IP addresses to router subinterfaces.

2. Lab Environment

Component	Description
Simulation Platform	Cisco Packet Tracer
Routers	Cisco 2911
Switches	Cisco 2960
End Devices	Department PCs, wireless clients, optional servers
Head Office Departments	Admin, HR, IT
Branch Office Departments	Sales, Support
Security Segment	Guest WiFi VLAN
Example Head Office Network	192.168.100.0/24
Example Branch Office Network	192.168.200.0/24

3. Project Objectives

- Design a complete office network with a Head Office and Branch Office.
- Use Cisco 2911 routers and Cisco 2960 switches.
- Create VLANs for multiple departments.
- Assign switch ports to the correct VLANs.
- Configure trunk links between switches and routers.
- Configure router-on-a-stick subinterfaces using 802.1Q encapsulation.
- Enable inter-VLAN routing for authorized internal communication.
- Add wireless access and guest WiFi segmentation.
- Use ACLs or access restrictions to prevent guest access to internal departments.
- Test and document connectivity between devices and networks.
- Troubleshoot configuration errors and document the resolution.

4. Network Topology Summary

The topology was designed as a two-site office network. The Head Office included Admin, HR, and IT departments. The Branch Office included Sales and Support departments. Each department was separated using VLANs, and each VLAN used a router subinterface as its default gateway.

Cisco 2960 switches provided access-layer connectivity for department PCs. Cisco 2911 routers provided routing between VLANs and between office sites. Wireless access was included for users, and a guest WiFi VLAN was planned to isolate untrusted guest devices from internal business systems.

5. VLAN and IP Addressing Plan

VLAN ID	Name	Location	Purpose	Example Network
10	Admin	Head Office	Administrative users	192.168.100.0/27
20	HR	Head Office	Human Resources users	192.168.100.32/27
30	IT	Head Office	IT support and administration	192.168.100.64/27
99	Management	Head Office	Network management	192.168.100.96/27
100	Guest WiFi	Head Office	Guest wireless access	192.168.100.128/27
40	Sales	Branch Office	Branch Sales users	192.168.200.0/27
50	Support	Branch Office	Branch Support users	192.168.200.32/27
199	Branch Mgmt	Branch Office	Branch network management	192.168.200.64/27

6. Work Performed

6.1 Topology Build

- Placed routers, switches, PCs, wireless access points, and optional server devices in Packet Tracer.
- Organized the topology into Head Office and Branch Office sections.
- Connected devices using appropriate copper straight-through or serial links depending on the device role.

6.2 VLAN Creation

- Created department VLANs on Cisco 2960 switches.
- Named VLANs according to department role.
- Verified VLAN configuration with show vlan brief.

6.3 Access and Trunk Port Configuration

- Assigned end-user PC switch ports to the correct department VLAN.
- Configured trunk links between switches and router-facing interfaces.
- Verified trunk status with show interfaces trunk.

6.4 Router-on-a-Stick Configuration

- Created router subinterfaces for each VLAN.
- Applied encapsulation dot1Q before assigning IP addresses.
- Used subinterface IP addresses as the default gateways for VLAN clients.

6.5 Inter-VLAN and Branch Routing

- Configured routing to allow authorized communication between department networks.
- Validated routing tables and interface status.
- Tested connectivity between Head Office and Branch Office clients.

6.6 Wireless and Guest Segmentation

- Added wireless access where required.
- Placed guest users into a separate guest VLAN.
- Designed restrictions so guest traffic could not access internal department networks.

6.7 ACL and Security Testing

- Planned or configured ACLs to restrict guest VLAN access to internal VLANs.
- Tested guest-to-internal connectivity to confirm restrictions.
- Verified authorized internal traffic still functioned.

6.8 Connectivity Validation

- Performed PC-to-gateway, same-VLAN, inter-VLAN, and branch-to-head-office ping tests.
- Used show commands to validate VLANs, trunks, routes, and ACLs.
- Documented successful and blocked traffic flows.

7. Network Testing Results

Test	Source	Destination	Expected Result
Gateway Ping	Admin PC	Admin VLAN Gateway	Success
Gateway Ping	HR PC	HR VLAN Gateway	Success
Gateway Ping	IT PC	IT VLAN Gateway	Success
Same VLAN Ping	Admin PC 1	Admin PC 2	Success
Inter-VLAN Ping	Admin PC	IT PC	Success if allowed
Branch Connectivity	Sales PC	Head Office Router	Success
Branch-to-HQ Ping	Support PC	Admin PC	Success if allowed
Guest Isolation	Guest WiFi PC	HR or Admin PC	Blocked

8. Key Verification Commands

Command	Purpose
show vlan brief	Verify VLAN creation and access port membership.
show interfaces trunk	Verify trunk links and allowed VLANs.
show ip interface brief	Verify router and switch interface status.
show running-config	Review current device configuration.
show ip route	Verify routing table and reachable networks.
show access-lists	Verify ACL entries and traffic restrictions.
ping	Test host, gateway, VLAN, and branch connectivity.

9. Issues Encountered and Resolutions

Issue	Cause	Resolution
Router subinterface IP error	IP address was attempted before VLAN encapsulation was configured.	Configured encapsulation dot1Q first, then applied the gateway IP address.
VLAN connectivity	Access ports were missing or	Reviewed show vlan brief and

failure	assigned to the wrong VLAN.	corrected switchport access VLAN assignments.
Trunk not carrying VLAN traffic	Trunk mode or allowed VLANs were not configured correctly.	Configured switchport mode trunk and verified with show interfaces trunk.
Branch-to-HQ failure	Routing or IP addressing was incomplete.	Reviewed interfaces, routes, and ping tests; corrected gateway or route configuration.
Guest reaching internal networks	Guest VLAN restriction was not applied or tested.	Applied ACL restrictions and validated blocked traffic to internal VLANs.

10. Skills Demonstrated

- Cisco Packet Tracer topology design.
- Cisco 2911 router configuration.
- Cisco 2960 switch configuration.
- VLAN creation and verification.
- Access and trunk port configuration.
- 802.1Q router-on-a-stick configuration.
- Inter-VLAN routing.
- IP addressing and subnet planning.
- Branch office connectivity.
- Wireless and guest network segmentation.
- ACL-based traffic restriction planning.
- Network testing and troubleshooting.
- Professional technical documentation.

11. Final Outcome

The final Packet Tracer Office Network successfully represented a small enterprise network with separate department VLANs, routed connectivity, Head Office and Branch Office communication, wireless support, guest segmentation, and documented testing. The project produced a portfolio-ready demonstration of core networking skills and Cisco configuration concepts.

The completed deliverables include a README.md file, final lab report PDF, recommended commands file, topology file placeholder, screenshots, IP addressing table, and testing results documentation.

12. Reflection

This project reinforced that successful network design depends on careful planning before configuration. VLANs, gateways, trunk links, router subinterfaces, IP addressing, and ACLs all depend on each other. The most important troubleshooting lesson was learning the correct order for router subinterface configuration: 802.1Q encapsulation must be configured before an IP address can be applied.

The project also showed the value of testing after each stage. Verifying VLANs, trunks, routing, gateways, and ACLs helped isolate problems quickly and produced documentation that can be used for a resume, GitHub portfolio, or job interview discussion.